

Peripheral Blood Immune Gene Dysregulation in Parkinson's Disease: A Computational Analysis Supporting the Gut-Brain Axis Hypothesis

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Abstract

Parkinson's disease (PD) is a type of neurodegenerative disease characterized by the loss of dopaminergic neurons and the buildup of alpha-synuclein. Due to neuronal loss that comes early on despite the diagnosis happening much later, peripheral blood biomarkers can help us to detect early systemic changes of PD in an easier way. Using three publicly available peripheral blood Gene Expression Omnibus (GEO) datasets (GSE22491, GSE72267 and GSE99039), this paper looked at the difference in gene expression levels between PD subjects with healthy controls. Differentially expressed genes were characterized using the results of functional enrichment analysis via Gene Ontology Biological Process, Reactome, and DisGeNET from Enrichr. Although only LRRN3 was the gene that could be matched at the protein level, the analysis still pointed out the common aspects in biological processes including immune inflammatory signaling, gut barrier and microbial-response pathways, mitochondrial and metabolic dysfunction. Apart from the LRRN3, PD-related genes such as those involving TLR4-mediated innate immunity pathway, complement system dysfunctions, and intestinal epithelial integrity appeared in all datasets. This study, in general, offers some insight that peripheral blood transcriptomes might reflect, in a way, the systemic changes of the disease which are linked to PD pathogenesis. In addition, even after the multiple testing was corrected for, several of the disease associations with DisGeNET remained significant which was not the case with the GO Biological Process and Reactome pathway enrichments. Because of this, these conclusions at a pathway level, although promising, are still to be checked through larger-scale and/or longitudinal studies with a multi-omic approach.